Teaching debugging

Programming · Level 3 · Lesson 3



Today we will

- Teach a debugging routine.
- Read Python’s error messages with a class.
- Use printing values and explaining aloud as tools.

Remember

- The error names the kind and the line: look there and just before.
- No error, wrong answer? Print the values.
- Ask, don’t fix.

Bug detectives

1. Warm-up (5 min): Read an error message aloud together.
2. Teach (10 min): The routine, and the four common errors.
3. Hunt (15 min): Pairs debug three programs on paper.
4. Share (5 min): Which step of the routine found each bug?
5. Wrap-up (5 min): Make the poster. Exit ticket.

Hunting bugs on paper works with no devices; run the fixed programs on one phone at the end.

Let’s discuss

1. Why is fixing a pupil’s bug for them a missed chance?
2. How would you teach a bug that gives no error?
3. How can we make bugs feel normal in our class?

Exit ticket

- Write the question you will ask the next pupil who says “it doesn’t work”.